Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/14/2020
PI Name:	Clark, Noel A. Ph.D.		
Project Title:	Ferromagnetic Liquid Crystal Colloids in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309-0001	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/02/2016	**End Date:**	12/01/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
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Flight Program:	ISS		
Flight Assignment:	Liquid Crystal Facility NOTE: End date changed to 12/01/2022 per NSSC information. Previous end date was 12/01/2021. (Ed., 12/2/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Glaser, Matthew Ph.D. (University of Colorado, Boulder) Maclennan, Joseph Ph.D. (University of Colorado, Boulder) Park, Cheol M.S. (University of Colorado, Boulder) Shuai, Min Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	NNX17AC74G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Paramagnetic ferrofluids are familiar as suspensions of magnetic particles in solvents that become strongly magnetized in applied fields. A longstanding challenge has been to make such fluids ferromagnetic, so that they exhibit spontaneous macroscopic ferromagnetic ordering even in the absence of applied field. Recently, ferromagnetic fluid phases have been achieved by the ferromagnetic orientation of magnetic nanoplates in colloidal suspensions, either by dispersion in a thermotropic nematic liquid crystal (LC) host or by spontaneous nematic ordering in an isotropic solvent. These novel materials are optically birefringent, dichroic, and translucent, so that structures and textures can easily be visualized in polarized light. They manifest a variety of interesting and distinctive magnetic interaction effects and, because of the static magnetization, display ultrahigh sensitivity to externally applied magnetic fields. Field-induced changes in the shape of fluid drops, such as interfacial magnetic spike instabilities, occur even in the Earth's magnetic field and readily achievable benchtop magnetic fields are expected to induce spectacular magnetofluidic responses. Ferromagnetic nematics also exhibit distinctive magnetic self-interactions, including liquid crystal textures of fluid magnetic domains arranged in closed flux loops, that in microgravity should strongly affect the shape of free-floating drops. Freely suspended smectic LC films in the form of bubbles, the LC geometry currently studied in OASIS (Observation and Analysis of Smectic Islands in Space), will be rendered ferromagnetic by doping with magnetic nanoplates and manipulated magnetically. In suspensions studied on Earth, the typically more dense liquid crystal phase sediments to the lower parts of test cells, leaving a sharp interface with the co-existing isotropic phase. Microgravity offers the opportunity to perform critical experiments that are not possible on Earth, such as the observation of ferromagnetic droplet and other fluid interface shapes as a function of applied magnetic field, investigations of magnetic convective instabilities and thermocapillary effects resulting from temperature gradients, studies of liquid crystal ordering kinetics in the absence of gravity, and magnetic islands on smectic bubbles. The proposed research has both fundamental and applied aspects. One of the most interesting scientific subthemes of ferromagnetism is ferrofluidics, the study and application of paramagnetic, colloidal suspensions of sub-micron size ferromagnetic particles dispersed in solvents with random orientation of the magnetic dipoles. In ferrofluids, originally developed by NASA for enabling transport of rocket propellant in space vehicles, these nano-magnets orient in applied fields, producing a bulk magnetization that in turn generates forces and torques on the host fluid. This results in a variety of exotic and useful magneto-mechanical effects, including field-induced transport and radical changes of shape, which have led to a wide variety of technical and biomedical applications. Ferromagnetic nematics combine the traditional advantages of liquid crystal ordering with permanent magnetization, leading to delicate temperature control of the intrinsic magnetic order and a facile response to applied magnetic fields that suggests a range of enhanced applications analogous to those of conventional ferrofluids. Experiments in microgravity will enable the investigation of the fundamental properties of this new family of colloidal materials and of physical phenomena that cannot easily be probed on Earth. Microgravity investigations will be carried out using the OASIS hardware in the Materials Science Glovebox on International Space Station (ISS) with various modifications. The experiments will use the OASIS high and low resolution video cameras in their orthogonal view geometry. Magnetic freely suspended smectic bubble experiments will employ a slightly modified OASIS sample chamber. The other experiments will require new sample box designs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Paramagnetic ferrofluids are familiar as suspensions of magnetic particles in solvents that become strongly magnetized in an applied field. A longstanding challenge has been to make such fluids ferromagnetic, so that they exhibit spontaneous macroscopic ferromagnetic ordering even in the absence of applied field. Recently, ferromagnetic fluid phases have been achieved by the ferromagnetic orientation of magnetic nanoplates in colloidal suspensions, either by dispersion in a thermotropic nematic liquid crystal host or by spontaneous nematic ordering in an isotropic solvent. These novel materials are optically birefringent, dichroic, and translucent, so that structures and textures can easily be visualized in polarized light.
Task Progress:	A colloidal solution of BF nanopates in BuOH was observed to phase separate into a bulk isotropic phase and a nematic phase that took the form of cylindrical filaments. The filaments were ferromagnetic and underwent a Fréedericksz transition if an external magnetic field that exceeded a certain threshold was applied antiparallel to the magnetization direction of the filaments. The transition was characterized by conversion of the cylindrical filament to an undulating, quasi-sinsusoidal structure. The threshold varied for different filaments, depending on their width: thin ones had higher thresholds relative to thick ones. The amplitude of the undulation increased with the strength of the applied field, until adjacent segments of the filament on either side of the undulation peaks converged to form droplets. Future work will be done to improve the measurements of field threshold as a function of filament width. Ideally, the experiments would be conducted in a zero-field environment so that background fields do not exert undue influence on the filaments, and the applied magnetic field can be calibrated reliably as a function of distance from the sample. We also plan to study the dynamics of the Fréedericksz transition and the influence of filament length on the propagation of the undulations.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Abstracts for Journals and Proceedings	Shuai M, Chen X, Park CS, Maclennan JE, Glaser MA, Clark NA. "Phase Separation of Ferromagnetic Nematic and Isotropic Colloidal Suspensions." Presented at the APS March Meeting 2019, Boston, MA, March 4-8, 2019. Bulletin of the American Physical Society. 2019 Mar;64(2): Abstract: F58.00003. http://meetings.aps.org/Meeting/MAR19/Session/F58.3 , Mar-2019
Abstracts for Journals and Proceedings	Shuai M, Chen X, Park CS, Maclennan JE, Glaser MA, Clark NA. "Droplets of ferromagneticnematiccolloidal liquid crystal." Presented at 17th International Conference on Ferroelectric Liquid Crystals, Boulder Colorado, August 4-7, 2019. Program and abstracts. 17th International Conference on Ferroelectric Liquid Crystals, Boulder Colorado, August 4-7, 2019. , Aug-2019

Abstracts for Journals and Proceedings	Shuai M, Chen X, Park CS, Maclennan JE, Glaser MA, Clark NA. "Ferromagnetic nematic colloidal LC droplets in isotropic background." Presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Shuai M, Qi Z, Park CS, Glaser MA, Maclennan JE, Glaser MA, Clark NA. "Ferromagnetic liquid crystal colloids under applied magnetic fields in microgravity." Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
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Articles in Peer-reviewed Journals	Klopp C, Trittel T, Eremin A, Harth K, Stannarius R, Park CS, Maclennan JE, Clark NA. "Structure and dynamics of a two-dimensional colloid of liquid droplets." Soft Matter. 2019 Oct 28;15(40):8156-63. https://doi.org/10.1039/C9SM01433K ; PMID: 31595938 , Oct-2019